

Clinical Case Bank

15 exam-style cases built from the *Integrated Clinical Revision Guide* (Ain Shams 2025–2026, Levels A–C)

How to use: each case is worth **10 marks**. Answer the case page on its own, then turn over: the next page has the model answer and the marking scheme. Doses are as printed in the source guide, so check them against your department's textbook.

#	Case	Specialty
1	Crushing chest pain with hypotension (inferior STEMI + RV infarct)	Cardiology
2	Child with flitting joint pains and a new murmur (rheumatic fever)	Cardiology
3	Breathless young woman who can only say single words (acute severe asthma)	Chest
4	Chronic cough, night sweats and an apical cavity (pulmonary TB)	Chest
5	Tired woman with heavy periods (iron-deficiency anemia)	Hematology
6	Macrocytic anemia with numb feet (B12 deficiency)	Hematology
7	Petechiae in a well young woman (ITP)	Hematology
8	Fever, neck stiffness and a rash (meningococcal meningitis)	Tropical / ID
9	Undulant fever in a farmer (brucellosis)	Tropical / ID
10	Collapse after an antibiotic injection (anaphylaxis)	Immunology
11	Blisters and sore lips after a new drug (SJS/TEN)	Immunology
12	Painless penile ulcer (primary syphilis)	Dermatology / STI
13	Itchy family at night (scabies)	Dermatology
14	"They'd be better off without me" (major depression + suicide risk)	Psychiatry
15	Confused and shaking on day 3 after surgery (delirium tremens)	Psychiatry

Case 1

Cardiology · 10 marks

A 58-year-old male smoker has had 2 hours of crushing retrosternal pain with sweating. BP 85/50 mmHg, pulse 54/min, JVP raised, lungs clear. ECG: ST elevation in II, III and aVF. The nearest primary-PCI centre is 3 hours away.

1. What is the diagnosis? Name the likely culprit artery and the extra ECG lead you would request (and why). **[2]**
2. Mention two investigations apart from the ECG and what each adds. **[1]**
3. Write the immediate drug treatment with doses. **[3]**
4. Which reperfusion strategy is appropriate here, and why? List 3 **absolute** contraindications to it. **[2]**
5. **Special:** Which drugs must be avoided in this patient, and how would you treat his hypotension? **[2]**

Case 1 — Answers & marking scheme

Q	Model answer	Marks
1	Inferior STEMI complicated by right-ventricular infarction (hypotension, ↑JVP, clear lungs). Culprit: RCA (~85%; otherwise LCx). Request right-sided leads (V4R) : ST↑ in V4R confirms RV infarct (proximal RCA).	1+0.5+0.5
2	Any two: serial troponin I/T (rises 3–6 h, peak 36–48 h); echo (wall motion, RV dilatation, mechanical complications); CBC, kidney function, electrolytes, lipids. Do not delay reperfusion waiting for results.	0.5 each
3	• Aspirin 162–325 mg chewed, then 75–100 mg daily • Clopidogrel 300–600 mg load → 75 mg daily (or Ticagrelor 180 mg → 90 mg bid) • Anticoagulant: Enoxaparin 30 mg IV bolus then 1 mg/kg SC q12h (<75 y), or UFH 60 U/kg IV (max 4000) → 12 U/kg/h • Atorvastatin 80 mg; O₂ only if hypoxic; Morphine 2–4 mg IV for refractory pain • Bradycardia: Atropine 0.5–1 mg IV (max 3 mg). Withhold β-blocker (shock, bradycardia).	0.5 each (max 3)
4	Fibrinolysis (PCI is >120 min away): door-to-needle ≤30 min, e.g. Tenecteplase (weight-based) or Alteplase 15 mg → 50 mg/30 min → 35 mg/60 min, or Streptokinase 1.5 MIU/60 min; then transfer for angiography. Absolute CI (any 3): prior intracranial hemorrhage · cerebral AVM · intracranial malignancy · ischemic stroke <3 months · suspected aortic dissection · active bleeding/bleeding diathesis · significant head/face trauma <3 months.	1 + 1
5	RV infarct is preload-dependent: avoid nitrates and diuretics (also PDE-5 inhibitors with nitrates). Treat hypotension with IV normal-saline boluses ; inotropes if no response.	1 + 1

Pearl: Inferior STEMI + hypotension + clear lungs = RV infarct until proven otherwise. Giving nitroglycerin here can crash the BP.

Case 2

Cardiology · 10 marks

A 12-year-old girl (35 kg) from rural Upper Egypt had a sore throat 3 weeks ago. She now has fever 38.4 °C and painful swollen knees, then elbows, each lasting a few days. There is a new apical pansystolic murmur radiating to the axilla. ESR 75 mm/h, CRP 5 mg/dL, ECG PR interval 220 ms.

1. What is the diagnosis? List the criteria she fulfils. **[3]**
2. Which investigations prove a recent streptococcal infection, and which assess the heart? **[2]**
3. Outline her acute treatment with doses (eradication + anti-inflammatory). Echo shows moderate MR. **[3]**
4. **Special:** Give the secondary prophylaxis regimen and how long it should continue. **[2]**

Case 2 — Answers & marking scheme

Q	Model answer	Marks
1	Acute rheumatic fever (first attack) with carditis. Jones criteria (moderate/high-risk population): Major: carditis (new MR murmur) + migratory polyarthriti s. Minor: fever $\geq 38^{\circ}\text{C}$, ESR ≥ 30 / CRP ≥ 3 (a long PR does not count as minor when carditis is already a major criterion). Plus evidence of preceding GAS infection → 2 major criteria = diagnostic.	1 + 1 + 1
2	GAS evidence: throat culture, rapid antigen test, rising ASO titre (anti-DNase B). Heart: echocardiography (valve lesion, effusion, LV function), ECG, CXR (cardiomegaly).	1 + 1
3	- Eradicate GAS: Benzathine penicillin G 1.2 MU IM once (she is >27 kg) or Penicillin V 500 mg bid ×10 d; allergy: Erythromycin 250 mg qid ×10 d. - Moderate–severe carditis: Prednisone 1–2 mg/kg/day (max 60 mg) for 2–4 weeks, then taper while adding aspirin. (Arthritis only would be Aspirin 80–100 mg/kg/day in 4 doses for 4–6 weeks.) Bed rest, treat heart failure if present.	1 + 1.5 + 0.5
4	Benzathine penicillin 1.2 MU IM every 3–4 weeks (or Penicillin V 250 mg bid; Sulfadiazine 1 g daily; allergy Erythromycin 250 mg bid). Duration: carditis with persistent valve disease → 10 years or until age 40 (whichever longer, often lifelong). If MR resolves → 10 years or until 21.	1 + 1

Pearl: Prophylaxis durations: no carditis 5 y / until 21 · carditis, no residual disease 10 y / until 21 · residual valve disease 10 y / until 40.

Case 3

Chest · 10 marks

A 24-year-old woman with asthma, who uses only a salbutamol inhaler, arrives breathless. She can speak only single words, is using accessory muscles, RR 34/min, pulse 128/min, SpO₂ 88% on air, PEF 40% of predicted, loud wheeze.

1. Classify the severity of this attack, giving 2 features that support it. **[2]**
2. Write the immediate management in the ED with doses. **[3]**
3. One hour later she is drowsy with a quiet chest. ABG: PaCO₂ 46 mmHg. Interpret this and state what you do next. **[2]**
4. How is asthma confirmed on spirometry when she is stable? **[1]**
5. **Special:** What is wrong with her current maintenance treatment, and what is the preferred GINA approach? **[2]**

Case 3 — Answers & marking scheme

Q	Model answer	Marks
1	Acute severe asthma. Any 2: single words · RR >30 · HR >120 · SpO ₂ <90% · PEF ≤50% · accessory muscles.	1 + 1
2	• O₂ to SpO₂ 93-95% • Salbutamol 2.5-5 mg nebulised (or 4-10 puffs via spacer) every 20 min for 1 h • Ipratropium 0.5 mg nebulised with salbutamol (×3 doses) • Prednisolone 40-50 mg PO ×5-7 d or Hydrocortisone 100 mg IV q6h • No response at 1 h → MgSO₄ 2 g IV over 20 min (aminophylline 5 mg/kg load reserved)	0.5 each + 1 for MgSO₄
3	Life-threatening asthma: a "normal"/raised PaCO ₂ (>42 mmHg) in a breathless asthmatic means she is tiring and arrest is near (silent chest, drowsiness). → ICU, prepare for intubation (↓consciousness, exhaustion, pH <7.30, PaCO ₂ >45).	1 + 1
4	FEV ₁ /FVC reduced (<0.75-0.80) with bronchodilator reversibility: FEV₁ ↑ >12% and >200 mL after 200-400 mcg salbutamol (or PEF variability >10%).	1
5	SABA alone (without ICS) is contraindicated. Preferred (Track 1): ICS-formoterol as reliever ± maintenance, stepped up by severity (after this attack at least step 3-4: low/medium-dose ICS-formoterol maintenance + reliever). Check inhaler technique and triggers.	1 + 1

Case 4

Chest · 10 marks

A 35-year-old man (60 kg) has had a cough for 5 weeks with streaky hemoptysis, evening fever, night sweats and 6 kg weight loss. CXR: right upper-lobe cavity.

1. List the investigations to confirm the diagnosis, saying what each offers. **[3]**
2. Write his treatment regimen with doses for his weight. **[3]**
3. Match the side effect to the drug: (a) orange urine, (b) red-green colour blindness, (c) acute gout, (d) burning feet. Which supplement prevents (d)? **[2]**
4. **Special:** Define treatment failure and MDR-TB. **[2]**

Case 4 — Answers & marking scheme

Q	Model answer	Marks
1	• Sputum ZN smear ×3 for acid-fast bacilli • GeneXpert MTB/RIF: result in 2 h and detects rifampicin resistance • Culture (LJ or MGIT) = gold standard; gives drug sensitivity • Also: HIV test, baseline LFT, visual acuity / colour vision (before ethambutol)	0.75 each
2	2 months HRZE then 4 months HR under DOTS (6 months total). For 60 kg: Isoniazid 300 mg (5 mg/kg, max 300) · Rifampicin 600 mg (10 mg/kg) · Pyrazinamide 1.5 g (25 mg/kg) · Ethambutol 900 mg (15 mg/kg) + Pyridoxine.	1 + 2
3	(a) Rifampicin · (b) Ethambutol (optic neuritis) · (c) Pyrazinamide (hyperuricemia) · (d) Isoniazid (neuropathy) → Pyridoxine 10-50 mg/day .	0.4 each
4	Treatment failure : smear still positive at ≥5 months of treatment. MDR-TB : resistant to at least isoniazid + rifampicin (needs 9-20 months of second-line therapy).	1 + 1

Pearl: Rifampicin is a CYP450 inducer, so warn about OCP failure and interactions with warfarin and ART. Streptomycin is contraindicated in pregnancy.

Case 5

Hematology · 10 marks

A 32-year-old woman (60 kg) with heavy periods complains of fatigue and craves ice. She has spoon-shaped nails and angular cheilitis. Hb 8 g/dL, MCV 68 fL, RBC $3.9 \times 10^{12}/L$, RDW high; the smear shows pencil cells.

1. Give 2 differential diagnoses of microcytic anemia and show how iron studies separate them. **[3]**
2. Calculate the Mentzer index and interpret it. **[1]**
3. Write her oral iron treatment (drug, dose, duration, how to take it) and how you check that she is responding. **[3]**
4. **Special:** Give 2 indications for parenteral iron and calculate her total iron deficit (Ganzoni), target Hb 15 g/dL. **[3]**

Case 5 — Answers & marking scheme

Q	Model answer				Marks
1		Iron deficiency	Chronic disease	β-thal trait	1.5 each
	Ferritin	↓↓ (<15–30)	N / ↑	N–↑	
	TIBC	↑ (>360)	↓ (<240)	N	
	TSAT	<16%	15–20% / N	N	
	Other	RDW ↑	marrow iron present	HbA2 >3.5%, target cells	
	Any two conditions correctly contrasted.				
2	MCV ÷ RBC = $68 \div 3.9 \approx 17.4$ (>13) → favours iron deficiency (<13 suggests thalassemia trait).				1
3	Ferrous sulfate 200–325 mg three times daily (65 mg elemental iron/tab), on an empty stomach 1 h before meals with vitamin C. Continue 3–6 months after Hb normalises to refill stores. Response: reticulocytes peak on days 7–10 , Hb rises 1–2 g/dL every 2–3 weeks. Also treat the cause (menorrhagia → gynaecology).				1 + 1 + 1
4	Indications (any 2): intolerance · poor compliance · malabsorption (e.g. celiac) · ongoing bleeding exceeding oral replacement. Ganzoni = wt × (target – actual Hb) × 2.4 + 500 = 60 × (15 – 8) × 2.4 + 500 = 1008 + 500 = ≈1500 mg .				1 + 2

Pearl: Ferritin is an acute-phase reactant. A level of 30–100 can still mean iron deficiency when there is inflammation.

Case 6

Hematology · 10 marks

A 55-year-old woman with hypothyroidism has fatigue, a lemon-yellow tint and tingling in both feet. Examination: loss of vibration sense and joint-position sense in the legs, positive Romberg test. Hb 8 g/dL, MCV 115 fL, WBC and platelets mildly low, reticulocytes low, LDH raised.

1. What is the most likely diagnosis and cause? Name the neurological complication. **[2]**
2. List the investigations that confirm it, including one that distinguishes it from folate deficiency. **[3]**
3. Write the treatment with doses for this patient. **[3]**
4. **Special:** A junior doctor wants to start folic acid while the results are pending. Why is this dangerous? **[2]**

Case 6 — Answers & marking scheme

Q	Model answer	Marks
1	Vitamin B12 deficiency megaloblastic anemia due to pernicious anemia (autoimmune; linked to thyroid disease). Neuro: subacute combined degeneration of the cord .	1 + 1
2	• Smear: macro-ovalocytes, hypersegmented neutrophils (≥ 5 lobes), pancytopenia, low retics • Serum B12 ↓; ↑ methylmalonic acid (MMA) + ↑ homocysteine (folate deficiency: MMA normal) • Anti-intrinsic factor / anti-parietal cell antibodies • Marrow (if done): hypercellular, megaloblastic	0.75 each
3	With neurological involvement: Hydroxocobalamin/B12 1000 µg IM on alternate days for 3 weeks , then every 3 months for life . (Without neuro: 1000 µg IM daily ×7–14 d → weekly ×4–8 wk → every 3 months.) Watch for hypokalemia early in treatment. Retic response on days 3–7.	2 + 1
4	Folic acid alone corrects the blood count but lets the neurological damage (SCD) progress irreversibly . Always check and replace B12 first.	2

Case 7

Hematology · 10 marks

A 24-year-old woman presents with petechiae on her legs, gum bleeding and epistaxis for 3 days. She is otherwise well and has no splenomegaly or lymphadenopathy. Platelets $8 \times 10^9/L$ (large platelets on the smear); Hb, WBC, PT and aPTT normal.

1. What is the most likely diagnosis? Give 3 conditions you must exclude. **[2]**
2. What do the bleeding time, PT and aPTT show, and how does this differ from hemophilia and DIC? **[2]**
3. Write her emergency management with doses. **[3]**
4. When would you transfuse platelets in this condition? **[1]**
5. **Special:** She relapses on tapering steroids. List the second-line options. **[2]**

Case 7 — Answers & marking scheme

Q	Model answer	Marks
1	Immune thrombocytopenia (ITP): IgG anti-GPIIb/IIIa. Exclude (any 3): pseudothrombocytopenia (EDTA clumping → repeat in citrate) · TTP (schistocytes) · leukemia/aplastic anemia · drugs · HIV/HCV · SLE.	1 + 1
2	ITP: platelets ↓, BT ↑, PT and aPTT normal . Hemophilia: normal platelets and BT, isolated ↑aPTT . DIC: everything abnormal (↓plt, ↑PT, ↑aPTT, ↑TT, ↓fibrinogen, ↑D-dimer, schistocytes).	1 + 1
3	Platelets <10 ×10 ⁹ /L with bleeding = emergency: IV methylprednisolone + IVIG 1 g/kg/day ×2 days (or 0.4 g/kg ×5 d). Then Prednisone 1 mg/kg/day (or Dexamethasone 40 mg/day ×4 d), tapered over 4–6 weeks. Avoid NSAIDs and IM injections.	1.5 + 1.5
4	Only for life-threatening bleeding (e.g. intracranial), given with IVIG/steroids, because transfused platelets are destroyed quickly.	1
5	Eltrombopag 50 mg daily or Romiplostim weekly SC (TPO agonists) · Rituximab · Splenectomy (cures ~ ² / ₃ ; vaccinate first).	0.5 each (max 2)

Case 8

Tropical / ID · 10 marks

A 19-year-old university-dorm student has 12 hours of fever, a severe headache and vomiting. Neck stiffness; a petechial rash that does not blanch with the glass test; GCS 14; no focal signs, no papilledema. BP 100/60 mmHg.

1. What is the most likely diagnosis and organism? **[1]**
2. What should happen in the first 30 minutes? When is a CT needed before LP? **[3]**
3. Describe the expected CSF findings. **[2]**
4. Give the empirical antibiotics and the adjunct, with doses and timing. **[2]**
5. **Special:** His roommates are worried. What do you give them? **[2]**

Case 8 — Answers & marking scheme

Q	Model answer	Marks
1	Acute bacterial meningitis with meningococemia: <i>Neisseria meningitidis</i> .	1
2	Do NOT delay antibiotics: take blood cultures → dexamethasone + empirical IV antibiotics immediately , then LP. CT first only if: focal signs, papilledema, GCS ≤10, immunocompromised, seizures . (Kernig/Brudzinski are specific but insensitive, so a negative sign does not exclude meningitis.)	1.5 + 1.5
3	Bacterial: opening pressure >250 mm H₂O ; WBC 1000-10 000/μL , PMN >80% ; glucose <40 mg/dL (CSF:blood <0.4) ; protein >100-500 mg/dL ; lactate >4 ; Gram stain: Gram-negative diplococci.	0.5 each (max 2)
4	Age 18-50: Ceftriaxone 2 g IV q12h + Vancomycin 15-20 mg/kg q8-12h (add Ampicillin 2 g q4h if >50 y / immunocompromised / neonate, for Listeria). Dexamethasone 10 mg IV q6h ×4 days , given 10-20 min before or with the first antibiotic dose.	1 + 1
5	Close-contact chemoprophylaxis (any one): Ciprofloxacin 500 mg once · Rifampin 600 mg bid ×2 days · Ceftriaxone 250 mg IM once . Also notify public health; meningococcal vaccine for dorm residents.	1.5 + 0.5

Case 9

Tropical / ID · 10 marks

A 40-year-old sheep farmer who eats home-made unpasteurised cheese has had 4 weeks of fever that comes and goes in waves, drenching sweats with a mouldy smell, low-back pain and a painful swollen right testis. There is mild hepatosplenomegaly.

1. What is the diagnosis? Name the complication he has. **[2]**
2. List the confirmatory investigations, including any special instruction to the lab. **[2]**
3. Write the preferred regimen with doses and duration, plus an all-oral alternative. **[3]**
4. MRI shows lumbar spondylitis. How does this change treatment? **[1]**
5. **Special:** His pregnant wife is also infected. What do you give her and why? What is the main cause of death in brucellosis? **[2]**

Case 9 — Answers & marking scheme

Q	Model answer	Marks
1	Brucellosis (<i>B. melitensis</i> , from sheep/goat dairy) complicated by epididymo-orchitis (back pain suggests sacroiliitis/spondylitis).	1 + 1
2	Blood / bone-marrow culture: warn the lab (prolonged incubation; lab-acquired infection risk). Serum agglutination test $\geq 1:160$. CBC, LFT.	1 + 1
3	Preferred: Doxycycline 100 mg bid $\times 6$ weeks + Streptomycin 1 g IM daily $\times 14-21$ days (relapse $< 5\%$). Oral alternative: Doxycycline 100 mg bid + Rifampin 600-900 mg daily $\times 6$ weeks (relapse $\sim 10\%$).	2 + 1
4	Complicated disease: Doxycycline + Rifampin + Ceftriaxone 2 g IV q12h (or streptomycin) for 12 weeks to 6 months .	1
5	Pregnancy: TMP-SMX + Rifampin $\times 6$ weeks , avoiding doxycycline (teeth/bone) and aminoglycosides (fetal ototoxicity). Main cause of death: endocarditis (aortic valve) , which needs valve replacement.	1 + 1

Case 10

Immunology · 10 marks

Five minutes after an IM penicillin injection, a 30-year-old woman develops generalised flushing and hives, a hoarse voice with stridor, wheeze and abdominal cramps. BP 70/40 mmHg, pulse 135/min, weak.

1. What is the diagnosis? Give 2 features that distinguish it from a vasovagal faint. **[2]**
2. Write the immediate management in order, with doses and route. **[4]**
3. Which blood test supports the diagnosis, and when should it be taken? **[1]**
4. **Special:** She improves after 30 minutes and wants to go home. What do you tell her, and what does she leave with? **[2]**
5. What is the immune mechanism? **[1]**

Case 10 — Answers & marking scheme

Q	Model answer	Marks
1	Anaphylaxis (anaphylactic shock). Vs vasovagal (any 2): tachycardia and a weak pulse (vasovagal: bradycardia) · urticaria/flushing/itch (vasovagal: pallor, no hives) · wheeze/stridor · shock that persists when supine.	1 + 1
2	Epinephrine 1:1000 IM, anterolateral thigh, 0.3-0.5 mg (0.01 mL/kg, max 1 mL); repeat every 5-15 min (up to 2 more doses). Never SC. - Lie flat with legs raised (never sit up) · 100% O₂ 8-10 L/min · large-bore IV · saline 1-2 L · monitor every 2-5 min. - Only after epinephrine: Hydrocortisone 250-500 mg IV, Diphenhydramine 25-50 mg + H2 blocker (famotidine), Salbutamol 2.5-5 mg neb for wheeze. - Refractory: IV epinephrine infusion; stridor/tongue edema → early intubation.	1.5 + 1 + 1 + 0.5
3	Serum tryptase within 1-2 hours of onset.	1
4	Keep her under observation for 6-24 h because of biphasic anaphylaxis (recurrence 3-4 h later, in up to 20%). On discharge: epinephrine auto-injector (0.3 mg) with training, a penicillin-allergy label and allergist referral.	1 + 1
5	Type I (IgE-mediated) hypersensitivity: allergen cross-links IgE on mast cells → degranulation (histamine, tryptase, leukotrienes).	1

Case 11

Immunology / Dermatology · 10 marks

A 45-year-old man started allopurinol 2 weeks ago. He now has fever, painful dusky-red macules that are blistering and peeling (about 15% of body surface), positive Nikolsky sign, hemorrhagic crusted lips, red eyes and genital erosions.

1. What is the diagnosis? How is it classified by body-surface area? **[2]**
2. Which hypersensitivity type is this? **[1]**
3. Outline management. **[4]**
4. After recovery, can a drug provocation test confirm the culprit? **[1]**
5. **Special:** How would DRESS differ clinically from this patient? **[2]**

Case 11 — Answers & marking scheme

Q	Model answer	Marks
1	SJS/TEN overlap (15% detached), drug-induced (allopurinol). Detached BSA: SJS <10% · overlap 10-30% · TEN >30% ; ≥2 mucosal sites involved.	1 + 1
2	Type IVc : CD8 cytotoxic T cells (perforin/granzyme, Fas-L), onset 4–28 days.	1
3	• Stop the drug immediately (and any non-essential drugs) • Admit to ICU/burn unit; barrier nursing, wound care • Fluids and electrolytes, nutrition, temperature control • Eye care (ophthalmology), mouth and genital care, lung care • IVIG or cyclosporine; SCORTEN to predict mortality	1 for stopping drug; 0.75 each other (max 4)
4	No : provocation is absolutely contraindicated after SJS/TEN (also DRESS, exfoliative dermatitis). The drug is avoided for life.	1
5	DRESS: later onset (2–6 weeks), fever + widespread rash + internal organ involvement (hepatitis, nephritis, pneumonitis, myocarditis), eosinophils >10% , atypical lymphocytes, lymphadenopathy; no extensive epidermal detachment. Type IVb.	2

Case 12

Venereology · 10 marks

A 25-year-old man has a single painless ulcer on the coronal sulcus that appeared 3 weeks after unprotected sex. The ulcer is firm and indurated with a clean base. He has bilateral rubbery, non-tender, non-suppurating inguinal nodes.

1. What is the most likely diagnosis? Give 2 differential diagnoses and how each differs. **[3]**
2. List the investigations and the role of each. **[3]**
3. Write the treatment, including the option if he is allergic to penicillin. **[2]**
4. **Special:** 6 hours after the injection he has fever, chills and tachycardia. What is happening, and what do you do? **[1]**
5. How do you confirm cure? **[1]**

Case 12 — Answers & marking scheme

Q	Model answer	Marks
1	Primary syphilis (hard chancre). Chancroid: 3–7 d incubation, multiple, very painful, soft ulcers with ragged undermined edges and purulent base; tender suppurative bubo. Genital herpes: multiple grouped painful vesicles → shallow erosions; tender bilateral nodes.	1 + 1 + 1
2	• Darkfield microscopy of chancre exudate: motile spirochetes (immediate diagnosis) • VDRL/RPR (non-treponemal): screening and monitoring • TPHA / FTA-ABS (treponemal): confirmation; stays positive for life (FTA-ABS IgM earliest) • Screen for other STIs including HIV	0.75 each
3	Benzathine penicillin G 2.4 MU IM single dose. Allergy: Doxycycline 100 mg bid ×14 days (Tetracycline 500 mg qid ×15 d; in pregnancy Erythromycin 500 mg qid ×15 d). Treat partners.	1 + 1
4	Jarisch-Herxheimer reaction (endotoxin release from dying spirochetes, 2–24 h), not an allergy. Continue penicillin; give antipyretics ± steroids.	1
5	4-fold fall in VDRL/RPR titre on serial testing (treponemal tests cannot be used to monitor).	1

Pearl: Late latent syphilis (>1 y) needs Benzathine penicillin 2.4 MU IM **weekly ×3** (allergy: Doxycycline ×30 d).

Case 13

Dermatology · 10 marks

A father, his pregnant wife (24 weeks) and their 6-week-old baby all have severe itching that is worse at night. The father has excoriated papules and S-shaped burrows in the finger webs and on the wrists, around the umbilicus and on the genitalia. The baby has lesions on the scalp, palms and soles.

1. What is the diagnosis? Give 2 features that support it. **[2]**
2. Write the father's treatment with application instructions. **[3]**
3. **Special:** Which treatment is safe for the pregnant mother and for the 6-week-old baby? Which drug is contraindicated in both? **[3]**
4. What general measures stop re-infestation? **[2]**

Case 13 — Answers & marking scheme

Q	Model answer	Marks
1	Scabies (<i>Sarcoptes scabiei</i>). Any 2: nocturnal itch · several family members affected · burrows (S-shaped, ending in a vesicle) · typical sites (finger webs, wrists, axillae, umbilicus, genitalia). The head is spared in adults but involved in infants.	1 + 1
2	Permethrin 5% cream from neck to toes , left on for 8-12 h (overnight) then washed off; repeat on day 8 . Alternatives: Benzyl benzoate 25%, Ivermectin 200 µg/kg orally repeated on day 8. Antihistamine for itch, which can last 2-4 weeks after cure.	1 + 1 + 1
3	Precipitated sulfur (10% adult / 5% child) nightly for 3-5 days is safe in pregnancy and infants <2 months ; Permethrin 2.5% for older infants (include scalp and face in infants). Ivermectin is contraindicated in pregnancy and in children <15 kg.	1 + 1 + 1
4	Treat all close contacts at the same time , even without symptoms. Wash clothes and bed linen at >60 °C , or seal them in a bag for 10 days.	1 + 1

Pearl: Crusted (Norwegian) scabies occurs in immunocompromised patients and is highly contagious.

Case 14

Psychiatry · 10 marks

A 62-year-old man, widowed 6 months ago, has had 2 months of low mood, loss of interest, early-morning waking, 7 kg weight loss, poor concentration and guilt. He says his children "would be better off without me" and admits he has been saving his sleeping tablets. He has no history of mania. He had an MI last year and takes a β -blocker.

1. What is the diagnosis? Justify it using DSM-5 criteria. **[2]**
2. Which medical and drug causes should be excluded, and with which tests? **[2]**
3. **Special:** List his suicide risk factors and the immediate safety steps. **[3]**
4. Give the drug treatment with dose, and state when ECT is indicated. **[3]**

Case 14 — Answers & marking scheme

Q	Model answer	Marks
1	Major depressive disorder (severe) with suicidal ideation. ≥5 of 9 symptoms for ≥2 weeks, including depressed mood or anhedonia: mood, anhedonia, sleep, weight, concentration, guilt, suicidal thoughts. Duration and severity exceed normal grief.	1 + 1
2	Exclude: hypothyroidism (TSH) , diabetes (glucose), post-MI/stroke depression, anemia (CBC), B12; drugs: antihypertensives (β-blocker) , steroids , alcohol/substances. Ask about past mania or hypomania (bipolar: avoid antidepressant monotherapy).	1 + 1
3	Risk factors (any 4): male, elderly, widowed, severe depression, hopelessness, a plan with stockpiled pills , chronic illness. Actions: take every threat seriously ; calm, private interview + collateral history; remove means (pills, sharps, belts); 1:1 observation / admission ; document every step.	1.5 + 1.5
4	SSRI: Sertraline 50-200 mg/day (safest after MI), or Fluoxetine 20 mg / Citalopram 20-60 mg; + psychotherapy (CBT). Response lag 2-4 weeks; review often. ECT (6-12 sessions, 3/week): severe suicidality, stupor, food refusal, psychotic features, treatment resistance .	1.5 + 1.5

Case 15

Psychiatry · 10 marks

On day 3 after a hernia repair, a 48-year-old man becomes acutely confused and agitated, sees insects on the walls, and has a coarse tremor, sweating, temperature 38.2 °C, pulse 124/min and BP 165/100 mmHg. His wife says he drinks a bottle of spirits a day. He had a brief fit yesterday.

1. What is the diagnosis? Explain how the timeline fits. **[2]**
2. What features show this is delirium rather than dementia? Name 3 investigations. **[2]**
3. Write the management with doses. **Special:** what must be given before any glucose, and why? **[4]**
4. Name the triad of Wernicke encephalopathy and 2 drugs for relapse prevention. **[2]**

Case 15 — Answers & marking scheme

Q	Model answer	Marks
1	Alcohol withdrawal → delirium tremens (mortality 10–20%). Timeline from the last drink: 6–12 h tremor/anxiety/sweating → 12–48 h seizures → 72–96 h DTs (clouded consciousness, vivid visual hallucinations, autonomic hyperactivity).	1 + 1
2	Delirium: acute onset over hours, fluctuating, clouded consciousness / impaired attention , worse at night (dementia: insidious, clear consciousness until late). Tests (any 3): glucose, electrolytes incl. Mg and phosphate , LFT, CBC, infection screen, CT brain if head injury or focal signs.	1 + 1
3	• Thiamine 100 mg parenteral bid ×3–5 days, BEFORE any glucose: glucose without thiamine can precipitate Wernicke encephalopathy. Then oral thiamine for ≥6 months. • Benzodiazepine: Chlordiazepoxide 50–100 mg PO q4h or Lorazepam/Diazepam IV (BZDs are the treatment of choice here). • IV fluids, correct electrolytes (K, Mg), quiet lit room, reorientation, no restraints.	2 + 1.5 + 0.5
4	Wernicke: ataxia, confusion, ophthalmoplegia (Korsakoff = amnesia + confabulation). Relapse prevention (any 2): Naltrexone, Acamprosate, Disulfiram .	1 + 1

Pearl: In other kinds of delirium, avoid benzodiazepines (use Haloperidol 2–5 mg IM). Alcohol or sedative withdrawal is the exception.